SOMAIYA VIDYAVIHAR UNIVERSITY
K J Somaiya Institute of Management

Session 3 - Human Resource Planning

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Session Objectives

- LO1 Recognize and articulate the importance of Human Resource Planning (HRP) in achieving organizational goals
- LO2 Assess internal and external factors that influence HRP across diverse contexts
- LO3 Understand the HR Planning process and how it aligns with organizational/ businsess strategy
- LO4 Distinguish between approaches to manage gaps (between the demand and supply of human resources) and mitigate risks in varied organizational contexts.

https://economictimes.indiatimes.com/news/company/corporate-trends/bigge st-restructuring-tata-motors-shrinking-structure-of-its-white-collar-workforce/a rticleshow/56723362.cms

Tata Motors was losing its market share in the competitive and fast-growing automobile market of India and was Thrown at the sixth place in the passenger vehicle segment. After involvement of of some consulting firms , the Company's performance improves and they became 4th largest passenger vehicle company. Now the company aspires to improvise its ranking further and so in Jan 2017 , it announced the delayering of the organizational Structure by reducing the layers from 14 to 5. Top 2 levels would be the strategy executors. Though the plan was to be rolled out by April, the company has already hired 100 high performers for these levels. Organizational Restructuring was in process to reduce middle management positions and also multiple reporting. Tata Motors Selected 400 employees to be retained in these 5 levels out of which 100 for top 2 layers are chosen.

How did the company decide they would require 400 employees for these levels?
Who would the chosen employees be?
Why are they reducing the workforce and to what extent should it be done?

The answers lie in Manpower planning.

Human Resource Planning

Human Resource Planning (HRP) is the process to determine the way in which the business organizations should progress from its existing manpower position to the ideal position typically tailored to the needs and overall objectives of the business..

Fundamentally dynamic process that endeavours to monitor and manage the flow of people into, through and out of the organization.

It contains quantitative analyses of HR data ranging from head counts to costs to qualitative analyses about culture, learning and knowledge management.

Significance of Human Resource Planning (HRP)

Caselet – HRP in an Insurance Organization

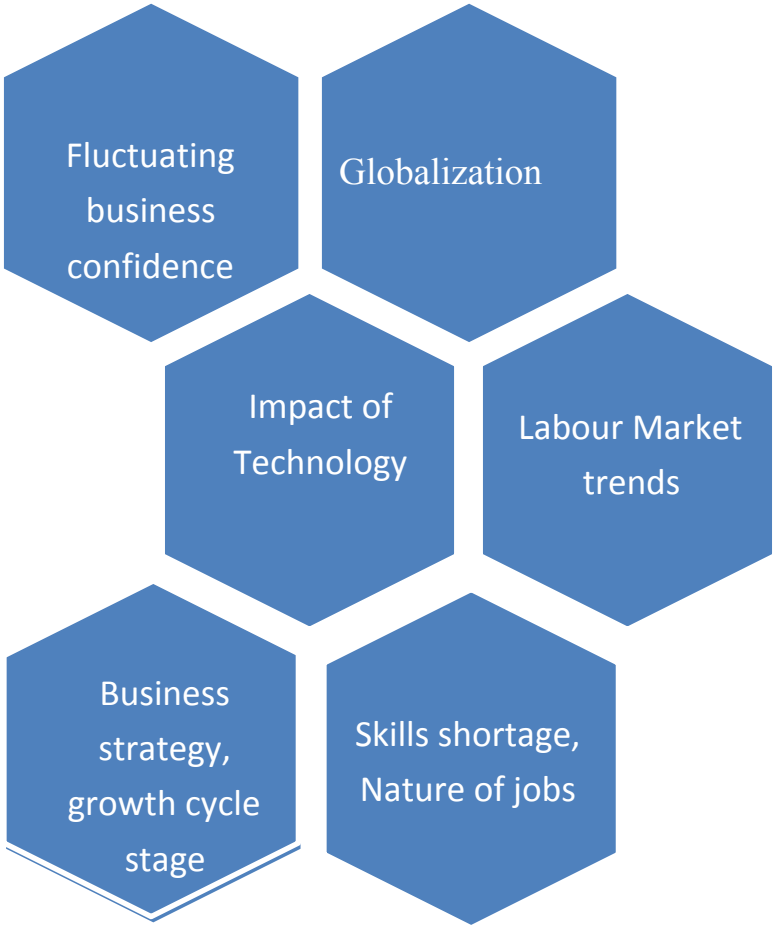
Factors affecting HRP

Caselet - HRP in TISCO

Significance of Human Resource Planning (HRP)

- Link HR plans with business plans
- Optimizes the use of resources
- Helps challenge manpower related assumptions
- Foundation of other HRM functions
- Conformity to government guidelines
- Talented manpower inventory

Factors affecting HRP

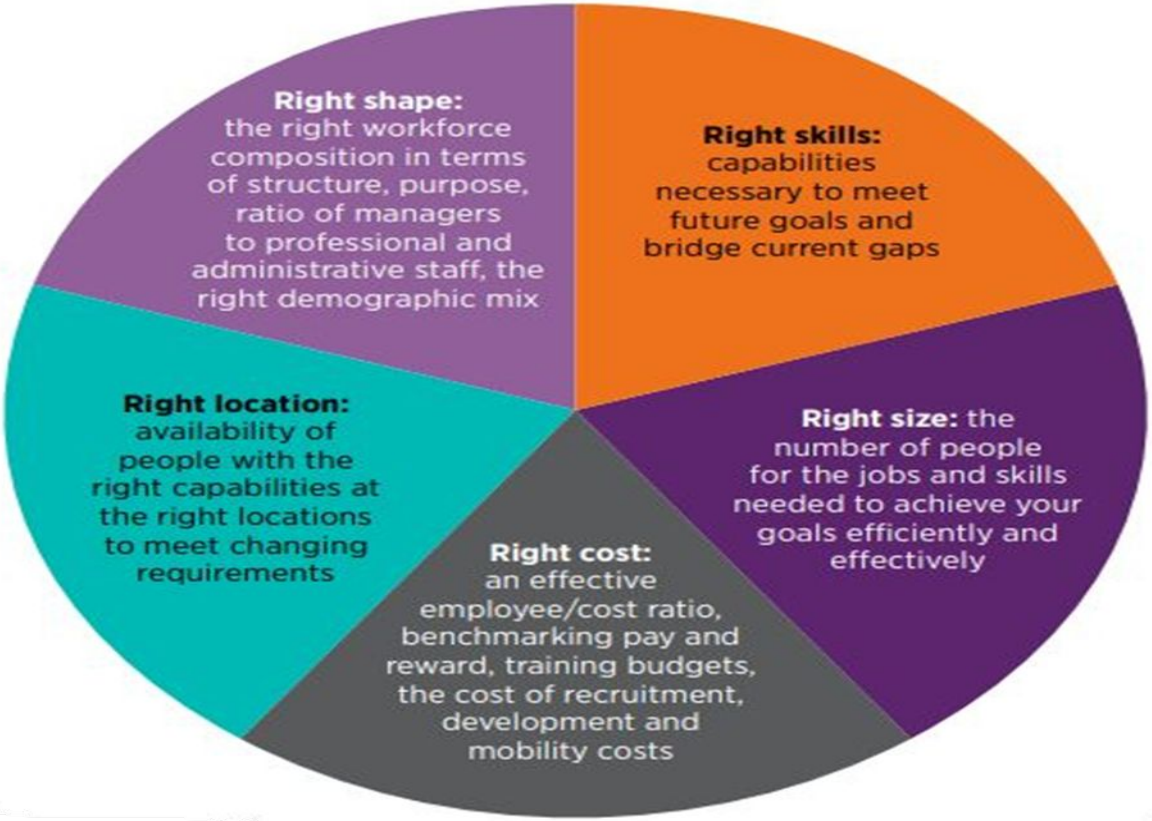


Introduction to Human Resource Planning (HRP)

- Human Resource Planning considers four factors:
  - Quantity. The number of employees needed by the company.
  - Quality. What skills, knowledge and abilities employees need to possess.
  - Space. The location where employees are needed
  - Time. When employees are required and for how long.
- It is a dynamic, iterative and interactive process, not subservient to corporate planning.

Introduction to Human Resource Planning (HRP)

- Human Resource Planning dimensions:



Introduction to Human Resource Planning (HRP)

- Human Resource Planning happens at various levels:
- At the national level, it is generally done by the government and covers various items like population projections, economic development programs, educational facilities, occupational distribution and growth, industrial and geographical mobility of human resources.
- At the sector level, it can be done by the Central and State Governments covering the manpower needs of various sectors such as industry, agriculture, service sector.
- At the industry level, it may cover manpower forecast for specific industries e.g., heavy industries, engineering, consumer goods industries, public utility industries, etc.
- At the industrial unit level i.e., micro level, it may cover its manpower needs for its various departments and for different types of personnel.

Thus, Manpower Planning plays an important role at both the micro as well as the macro level.

12 lakh manpower needed for semiconductor industry: Committee report

ETtech • Last Updated: Aug 04, 2023, 01:27:00 PM IST

FOLLOW US SHARE FONT SIZE SAVE PRI

Synopsis
The committee was constituted to develop a roadmap for "India as a Semiconductor Talent Nation", minister of state Rajeev Chandrasekhar said in a reply in the Lok Sabha.

India will need 12 lakh manpower in semiconductor-related industries by 2032, according to a report by the Semicon India Future Skills Talent Committee set up in August last year.

Read more at:
https://economictimes.indiatimes.com/tech/technology/12-lakh-manpower-needed-for-semiconductor-industry-committee-report/articleshow/102362667.cms?utm_source=contentofinterest&utm_medium=text&utm_campaign=cppst

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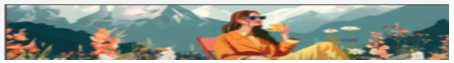


The government has taken an "ecosystem approach" and identified global standard talent development as a key requirement of this ecosystem, Chandrasekhar said. It has initiated a number of skilling and academic initiatives to develop the required capacities and capabilities in young Indians aiming for careers and opportunities in the electronics and semiconductor sector, he added.

The All India Council for Technical Education (AICTE) has formulated curricula for UG, diploma and minor degrees in the semiconductor domain in order to meet the industry's talent requirements, the minister said. Further, the Semicon India Future Skills/Chip to Startup (C2S) programme will create 85,000 highly qualified global talent over 5 years.

MoUs have also been signed between **India Semiconductor Mission** (ISM) and Purdue University, and the Indian Institute of Science, **Bengaluru**, and US-based semiconductor equipment manufacturer Lam Research for collaborations on R&D programmes and workforce training, respectively.

Introduction to Human Resource Planning (HRP)

- Human Resource Planning is also time based:

Time frame	Requirement basis	Availability	Possible action plans to meet requirements
Less than an year	Annual Budget (Operational Plans)	Existing manpower (by proper distribution of workload)+ contractual manpower	Body shoppers/ contractors, overtime, recruitment on contractual terms, restructuring and layoffs
1 – 2 years	Forecasted budget or forward budget (Business Plans)	Current manpower less projected attrition	Transfer, Promotion, new recruitment, restructuring, redundancy, T&D
2 – 5 years	Long – term Plans	Projected manpower (incl those who will be inducted during the period)	Succession Plans, recruitment, T&D, restructuring, redundancy
More than 5 years	Perspective Plans	Labour market , Education system	Succession Plans, management development, OD, redeployment , job restructuring

How is HRP organized and who does it?

At senior levels, both HR directors and board members should drive the strategic end of the process and set the agenda for workforce change.

Workforce planning needs strong links across an organization's functions and into strategic planning and finance. Involving some of these other stakeholders in the design of a workforce planning approach will help with implementation down the line.

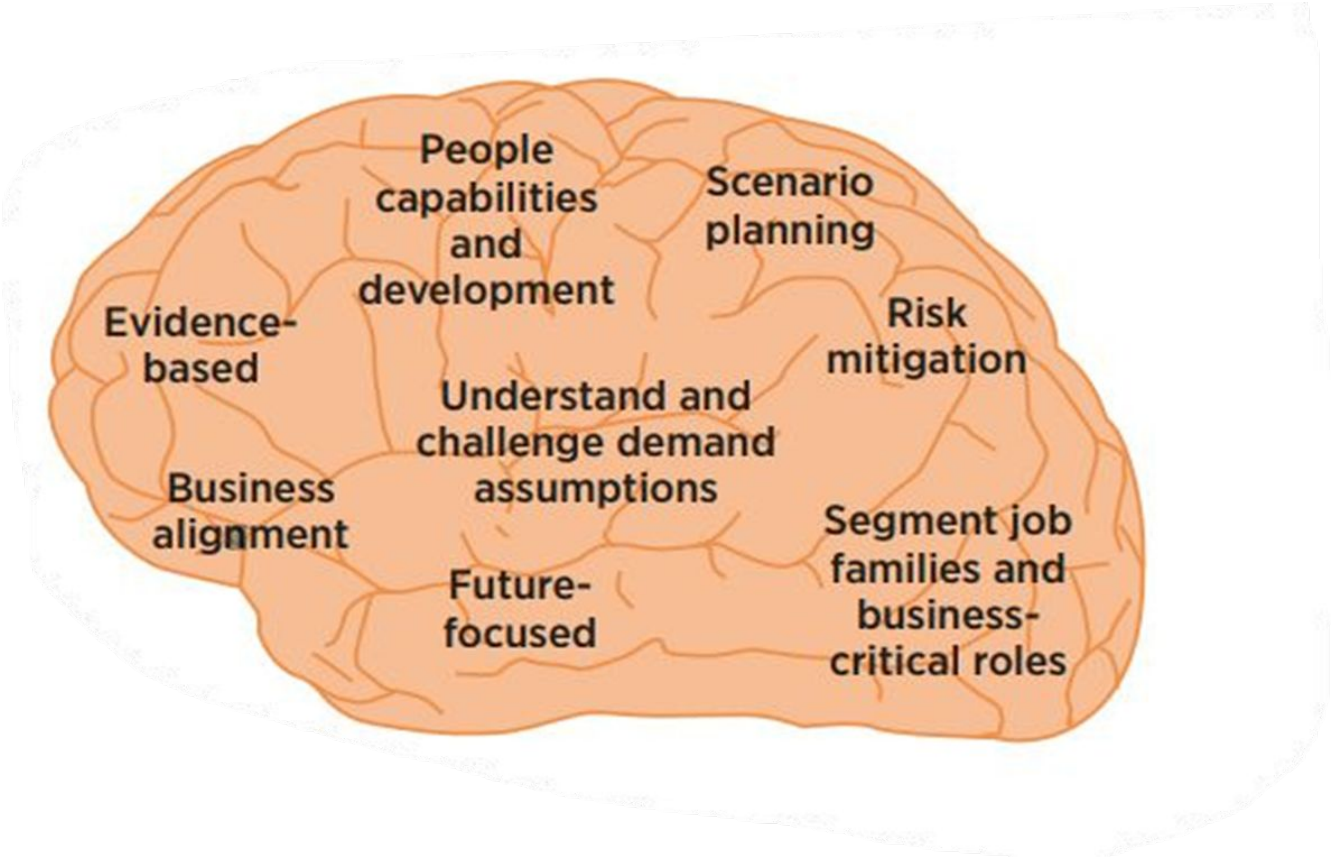
Longer-term planning and planning for key groups – such as senior managers – who are treated as a corporate resource tend to take place centrally.

The majority of workforce planning, takes place in the main business entities – divisions or regions of larger organizations.

Corporate functions often do their own planning for all their roles across the organization.

Managers frequently do their planning for operational employees at local unit level, as this is where employees are recruited, developed and deployed

Skills needed for HRP?



How does HRP work?

Manpower planning has been commonly described as a process consisting of three steps:

- predicting the future demand for manpower;
- predicting the future supply of manpower;
- looking at policies to reconcile any difference between the results of (a) and (b), often known as "closing the manpower gap".

A more detailed look at the three elements is as below:

- predicting demand may involve looking at productivity changes, technological changes, market forces and trends and the corporate strategy;
- predicting supply involves a knowledge of the current manpower stocks and looking at future recruitment, wastage, working conditions, promotion policies and labor market trends;
- closing the gap means examining training, remuneration, career planning, redundancies and further consideration of all the factors under the other headings.

How does HRP work?

How many people are we likely to need in our key operational and functional areas?

Do I have excess or surplus?

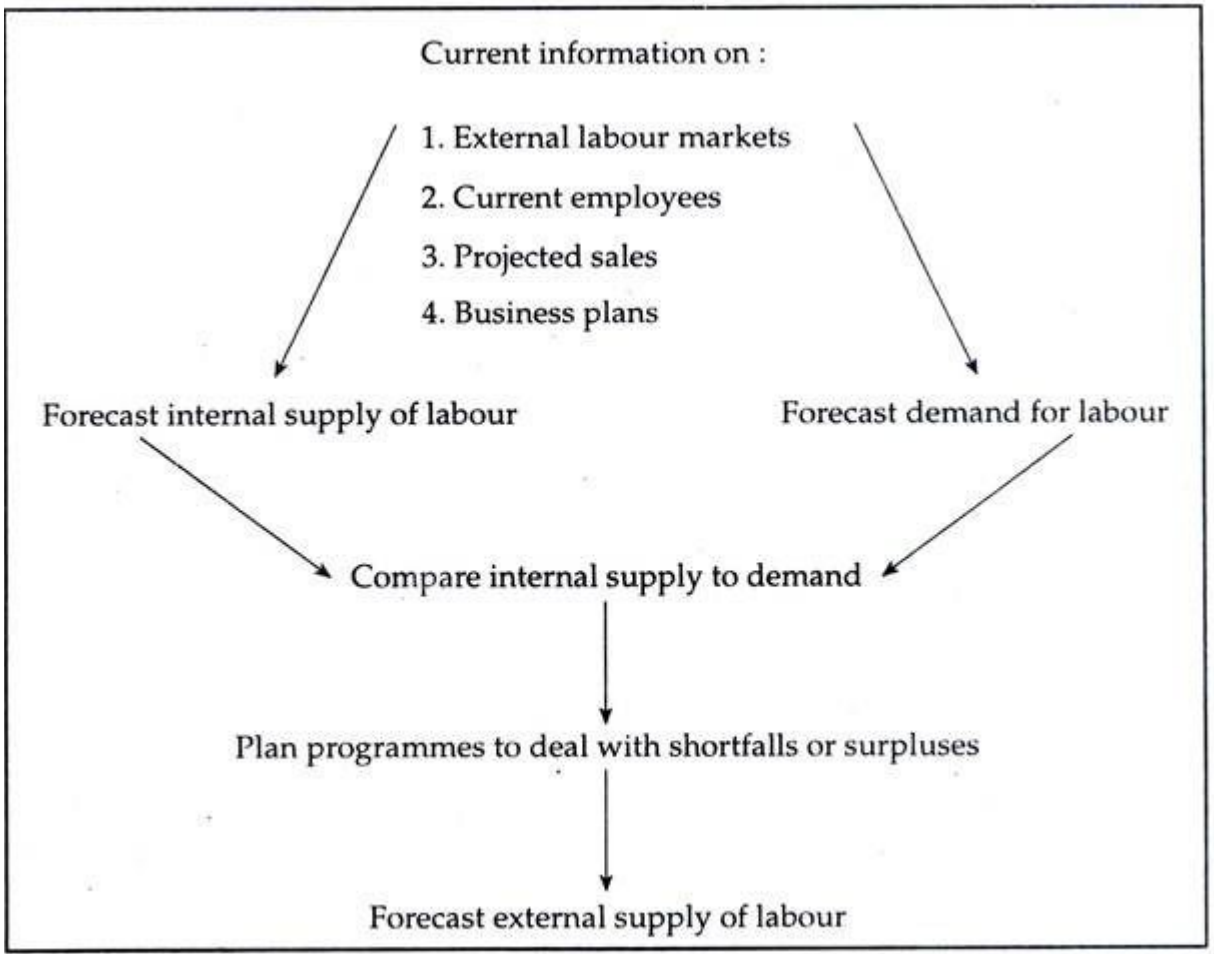
What skills are we likely to need in the future?

Will we be able to meet the needs from our existing resources?

If not, where will we be able to find them?

What do we need to do to develop or extend our skills base?

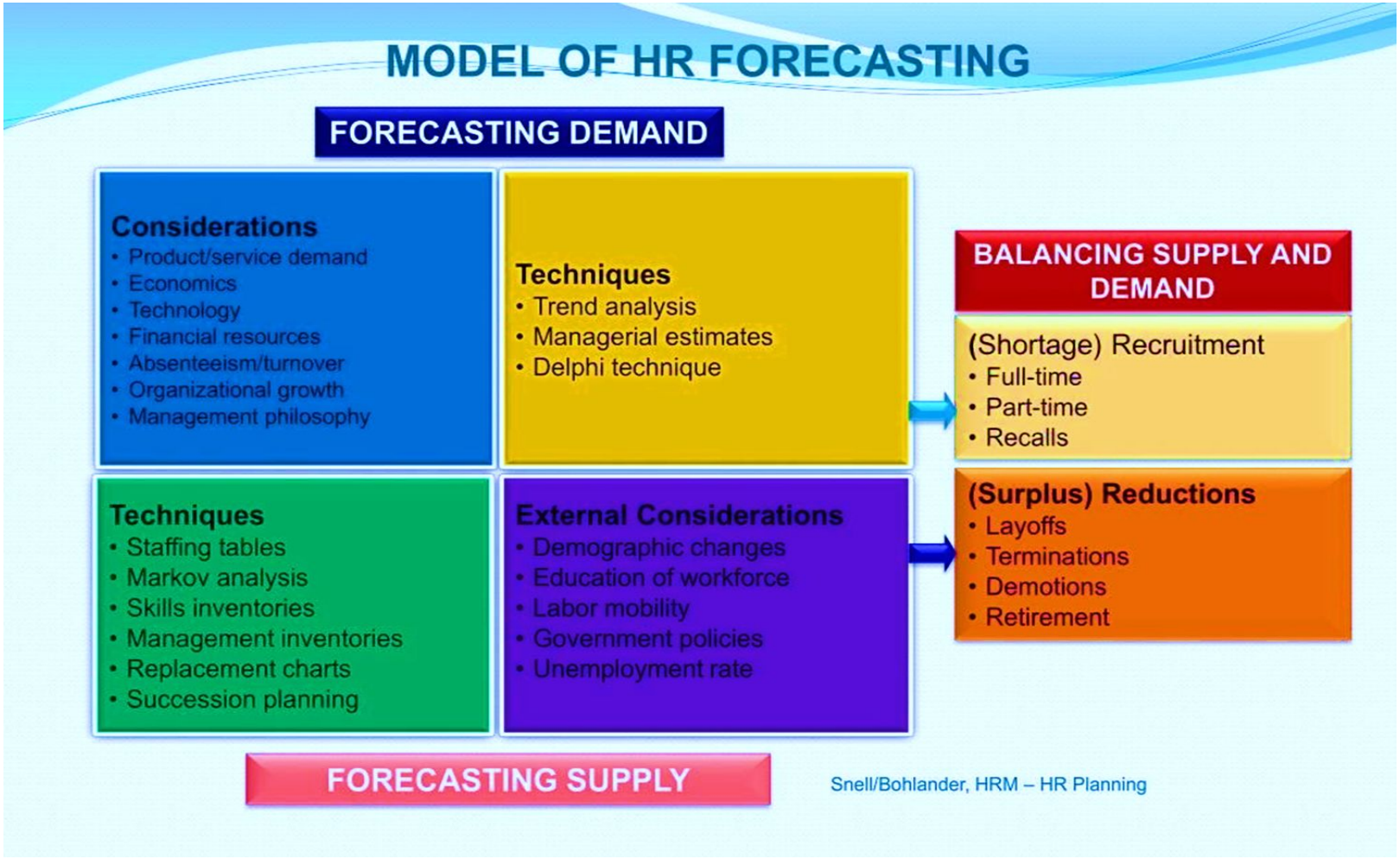
How does HRP work?



How does HRP work - Strategy

- Environmental Assessment – Scanning, Monitoring , Forecasting , Assessing
  - SWOT Analysis
  - PESTEL Analysis
- Current Staff Assessment
- Business Operating Model
- Organizational Strategy

How does HRP work – Demand and Supply Analysis



How does HRP work – Techniques for Demand Analysis

- Qualitative approaches
  - Managerial judgement – can be combined with quantitative tools. This can be done through both bottom-up forecasting approach and top-down forecasting approach.
  - Delphi technique – group process to achieve consensus forecast. No discussion within the panel of experts
  - Nominal group method – group process to achieve consensus forecast with discussion allowed within the panel of experts
  - Scenario Planning – provides multiple estimates of future HR demand, contingent on a unique set of assumptions and circumstances for each scenario.

How does HRP work – Techniques for Demand Analysis

Scenario Planning

Sample best- and worst-case scenarios

For a commercial bank that has recently added internet banking to its portfolio and is considering a digital-only platform:

Scenario 1	Scenario 2	Scenario 3
Customers embrace the digital age and disruptions – internet banking is the norm	Customers are reticent to embrace the digital-only banking solution and want a blended approach – the option of attending a branch to discuss financial options	New entrants enter the market. Big businesses that already have large databases and market a few specific banking products extremely well
If growth is an important driving force and the bank is considering expansion into another market:		
Do we have the in-house capability to redeploy expertise to that new account/function?	Is the best course of action to acquire a business that already has this expertise?	Would a partnership be the best course of action?

How does HRP work – Techniques for Demand Analysis

- Quantitative approaches
  - Time Series Analysis
  - Analysis of Productivity and Other ratios
  - Regression analysis
  - Work study techniques

How does HRP work – Techniques for Supply Analysis

- Internal Supply
  - Wastage Analysis- Indicates severance from the organization, which includes voluntary retirement, normal retirement, resignations, deaths and dismissals
  - Markov analysis – Flow model
  - Skill inventories/Management inventories
  - Replacement charts
  - Succession Planning
- External Supply - demographic changes, education of workforce, new graduates, labor mobility, government policies and unemployment rate

How does HRP work – Gap Closing Strategies

Labour Shortage		
Increase supply externally	Reduce demand	Improve retention
• **Recruit differently** • **Attract people of different age, sex, race, nationality, skill, qualifications and experience** • **Advertise in different places** • **Assess and select in different ways** • **Improve terms and conditions** • **Offer assistance in relocation, transport, childcare** • **Promote differently** • **Train existing staff** • **Redeploy existing staff**	• Redesign work • Redesign job specification • Use existing staff differently • Subcontract work • Relocate work • Automate	• Improve terms and conditions • Improvement of management style and communication • Improve promotion and recognition • Improve training • Improve inducements, bonus and the like • Improve selections, placements and inductions • Monitor staff turnover
Labour surplus		
Increase demand	Decrease supply	Discourage retention
• **Increase markets for products and services**	• Early retirement • Assisted career change • Secondments, sabbaticals, and the like • Voluntary redundancy • Compulsory redundancy	• Short term contracts • Flexible working

Example 1

- An HR manager has recently completed a thorough workforce analysis for the bank at which they work. The HR manager has discovered that a large percentage of the bank's mainframe programmers will retire over the next five years. The HR manager also discovered that the college graduates who have been recently hired with IT degrees did not specialize in mainframe technology. The HR manager recognizes the need to explore some creative ways to recruit new mainframe programmers in order to fill these upcoming vacancies.
- This is an example of gap analysis. By understanding the difference between the supply and demand for mainframe programmers, the HR manager is able to make informed decisions about how to address this issue.

Example 2

- An HR representative at an automotive manufacturing plant is planning recruitment strategies for the next year. The HR representative and their staff conduct a review of production and planning notes that they have received from the operation director. The information in the notes states that in the next production cycle advanced robots will replace a large group of employees who perform welding activities on an assembly line. The HR representative and their team decide to use this new information to alter their recruitment strategies. Additionally, they decide to give the opportunity for training for the welding employees so that they can potentially find positions in other departments.
- This is an example of demand analysis. The HR representative has identified a future need for employees with specific skills and has planned accordingly.

Identification of actionable areas

- 'Business-as-usual' action areas where recruitment, staff development and redeployment will be needed to keep additional needs in line with emerging changes in business requirements. It is helpful here to make a note of changes to skills needs which influence the more bottom–up process of individual development planning. It is also important to take account of estimates of likely losses when assessing likely gaps between available workforce and future needs.

- Workforce groups where recruitment, retention or both present resourcing challenges and alternative resourcing options may therefore need to be considered. These gaps may cover all jobs in a particular group or they may apply especially to some locations or teams.

- Workforce groups and/or parts of the business where workforce reductions may be necessary and which must be well managed.

- Change in the business strategy could lead to radical change in people and skills needs. Strategy change may be the result of competitive pressures, a new approach to product manufacturing or marketing, caused by emerging technology or the desire to reduce costs (as in offshoring).

- In extreme cases, resourcing difficulties may challenge the overall business strategy or prompt major areas of work for relocation or outsourcing. A combination of skills shortages, together with labour costs and different rates of market growth, may also drive companies to rebalance their global footprint towards different markets.

HR Plan Implementation

- Recruitment/Selection/Placement Plan
- Redeployment Plan
- Redundancy Plan
- Managerial Succession Planning
- Training Plan
- Productivity Plan
- Retention Plan
- Check / Review points

Calculating Staffing Needs

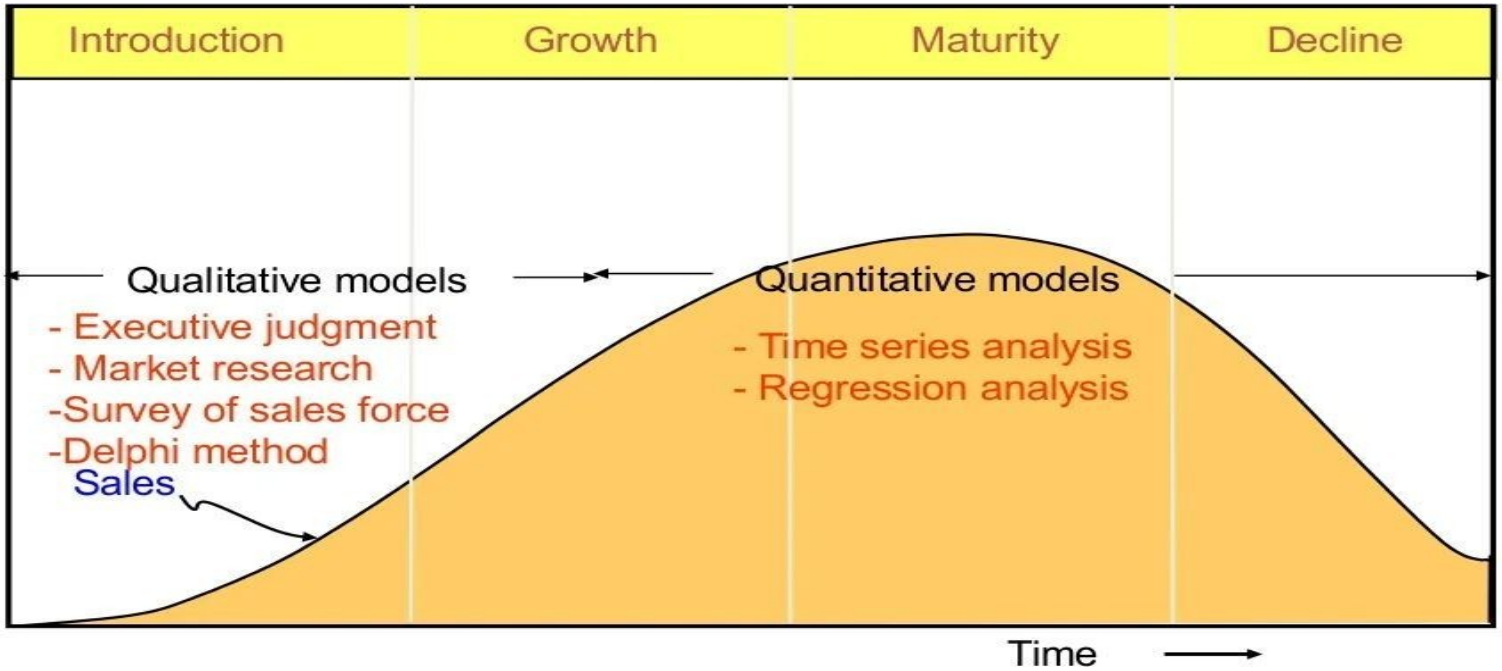
The formula is as follows:
Staffing Needs = (Workload / Productivity) x (1 + Buffer)

1. Workload: Measure your business's workload by assessing the tasks, projects, or customer interactions that need to be handled within a specific time frame. This can be done by analyzing historical data or using industry benchmarks.
2. Productivity: Evaluate the productivity of your current staff members to determine how much work they can handle within a given time period. Consider factors such as average response time, completion rates, or sales targets achieved. This will help you understand the productivity level per employee.
3. Buffer: The buffer factor accounts for unexpected events or changes in demand. It's essential to include a buffer to ensure that you have enough staff members to handle unforeseen circumstances without compromising productivity. The buffer can be a percentage based on historical data or industry standards.

By plugging in the values for workload, productivity, and buffer into the formula, you can calculate the staffing needs accurately. However, it's important to regularly reassess and adjust these values based on changes in your business environment.

Workforce group	Future staffing need	Current workforce	Resourcing issues, risks, gaps	Priority actions/options to mitigate risks
Marketing specialists	Numbers steady over planning period	• At least three will retire over next three years but voluntary turnover also expected to rise • Lack of good mid-career successors • Labour market extremely tight	• Need at least 10 new people over next 3 years • Significant risk of inability to recruit • Need to 'grow our own' more	• Support initial professional training for one or two people per year (start Year 2) • Accelerate development of less experienced professionals — aiming for three or four a year
Call centre operatives	Need more staff capable of dealing with difficult customers – current job role does not acknowledge this sufficiently	Already have a group of long-serving, experienced staff who could extend their technical and interpersonal skills	To date recruitment not a problem, but 15% of recruits have come from other countries. Could be an issue should immigration policy change	• Explore a higher-level job dealing with the most difficult customers – would need a more structured approach to bringing on talent • Explore knowledge transfer routes, for example coaching and mentoring by internal experts for internal staff • Moving some call centre work to other locations or recruiting from customer services in other sectors • Investigate using customer service apprenticeships
Senior leaders	Increasing need to lead people through a period of uncertainty	• Some will not like culture change and leave • Some high-performers lower down the organisation	Danger taking too long to raise our game as short-term talent pool weak	• Selective head-hunting, not necessarily from own sector • Development for existing senior managers on engaging employees • Targeted development for junior managers who could progress quickly

Forecasting During the Life Cycle



Prevalence of techniques

Techniques	% of companies using the technique
Supervisor estimates	69.7
Succession Planning or Replacement Charts	65.2
Rule of thumb or Non statistical formulas	37.9
Computer Simulation	18.2
Trend extrapolation	12.1
Regression Analysis	9.1
Delphi technique	3

Source – Charles Greer , SHRM , p184

Prerequisites to success of HRP

- There should be a proper linkage between HR plan and organizational plan.
- Top management support is essential.
- Proper balance should be kept between the qualitative and quantitative approaches to HRP.
- Involvement of operating managers is necessary.
- Proper alignment between short-term HR plans and long-term HR plans should be there.
- HR plan should have in-built flexibility in order to adopt to environmental uncertainties.
- Time period of HR plan should be appropriate to needs and circumstances of the organization.

Barriers to HRP

- Doubts over importance of having a future oriented HR practice
- Forecasting methods in use
- Doubts over HR's strategically aligned thinking
- Conflicts over short term vs long term plans
- Balance between quantitative and qualitative approaches
- Noninvolvement of operating/line managers

Class Assignment

There is a retail outlet with 50 sales personnel currently. According to the business strategy of the company , expansion plans will require the following manpower during the next 3 years:

a) 2025 - 60
b) 2026 - 70
c) 2027 – 80

There are no promotions into the group and typically 2 are promoted every year to the rank of sales team head per year. An average of 5 per year are transferred to other retail outlets of the organization and 2 per year are transferred in from other retail outlets. Employee turnover averages 10% of total sales personnel per year.

Q1 Calculate the recruitment needs of sales personnel for the said years
Q2 If we assume the average productivity of the sales personnel to be 75%, by how many numbers should the recruitment increase?

Q 1

Sales personnel at start	**50**			
		2025	**2026**	**2026**
Less:				
Promotions		2	3	4
Turnover		5	6	7
Total		7	9	11
Add				
Trf in		2	3	4
Total		2	3	4
Net		**45**	**54**	**63**
Expected employee nos		60	70	80
To be recruited		15	16	17

Q2: Increase in recruitment due to assumed productivity of 75%.

Assuming average productivity of sales personnel is 75%, it implies that each salesperson is effectively contributing 75% to the required output.

- Required output based on assumed productivity: $\frac{Required\,personnel}{0.75}$

Let's calculate the adjusted recruitment needs:

- For 2025: $\frac{15}{0.75} = 20$
- For 2026: $\frac{16}{0.75} \approx 21.33$ (round up to 22)
- For 2027: $\frac{17}{0.75} \approx 22.67$ (round up to 23)

Therefore, the recruitment should increase by approximately:

- 2025: $20 - 15 = 5$
- 2026: $22 - 16 = 6$
- 2027: $23 - 17 = 6$

So, the recruitment should increase by 5 in 2025, and by 6 each in 2026 and 2027, considering the assumed productivity of 75%.

Methods of Wastage Analysis

- ***Labour turnover index / Employee turnover rate*** indicates the number of employees leaving the organization as percentage to average number of employees.
- The following table exhibits manpower related information for a particular month from the company's HR files.
- Consider the date and calculate the voluntary and involuntary turnover index

Case 1: Calculate the Voluntary turnover (Attrition) rate

No. of employees at start of month – 500

Deaths – 2

Retirement due to accidents (employee not able to work) – 2

Resignations – 5

Dismissals – 1

Case 1: Calculate the Involuntary turnover rate

Avg employees during the period – 300

Retirements – 4

Layoffs – 10

Dismissals - 3

Voluntary turnover rate/index	(No of employees leaving the organization /Average number employees during the month)

Involuntary turnover rate/index	(No of employees leaving the organization /Total number of employees(period average)

$$\text{Hence average manpower wastage} = \frac{\text{Manpower leaving in a year} * 100}{\text{Average manpower in position}}$$

Table-3.13: Voluntary Turnover Index

Criteria	Number
Deaths	2
Accidents(Employee no longer able to do the work)	2
Resignations	5
Dismissals	1
Total number of employees leaving the organization during the month	2+2+5+1=10
Total number of employees remaining in the organization at the end of the month	500-10=490
Voluntary turnover rate/index	(No of employees leaving the organization /Average number employees during the month) $= \frac{5 * 100}{\frac{(500 + 490)}{2}}$ = 1.01% per month or 12.12 % per year

Table-3.14: Involuntary Turnover Index

Criteria	Number
Average number of employees during the period	300
Number of dismissals	3
Number of retirements through disability	4
Number of layoffs	10
Total number of employees leaving the organization during the month	3+4+10=17
Involuntary turnover rate/index	(No of employees leaving the organization /Total number of employees(period average) $= \frac{17 * 100}{300}$ = 6% during the period

Hence average manpower wastage = $\frac{\text{Manpower leaving in a year} * 100}{\text{Average manpower in position}}$

Thank You

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